

# Multimodal Product Matching

**FINALE**

Combining character-level TF-IDF and CLIP image embeddings  
on the Shopee Product Matching dataset

**99.00%**

Best held-out F1

**34,250**

Product listings

**22,028**

Balanced pairs

## Executive Summary

This report documents a multimodal product matching system built on the **Shopee Product Matching** dataset. The task is binary pair classification: given two product listings, each with a title and an image, decide whether they describe the same product. The system fuses **character-level TF-IDF title similarity** with **CLIP image-embedding similarity** and compares three ways of combining the two signals.

**99.00%**

Held-out F1 (learned fusion)

**99.39%**

Precision

**98.61%**

Recall

**+0.65 pp**

Gain over text-only

**Headline finding.** Text is the dominant signal (98.35% F1 alone), but a Logistic Regression that learns how to combine text and image similarity reaches **99.00% F1** on an untouched test set. Image information is substantially weaker alone (91.20% F1) yet adds complementary evidence. Forcing both modalities to agree with a hard AND rule does **not** help; treating images as soft evidence does.

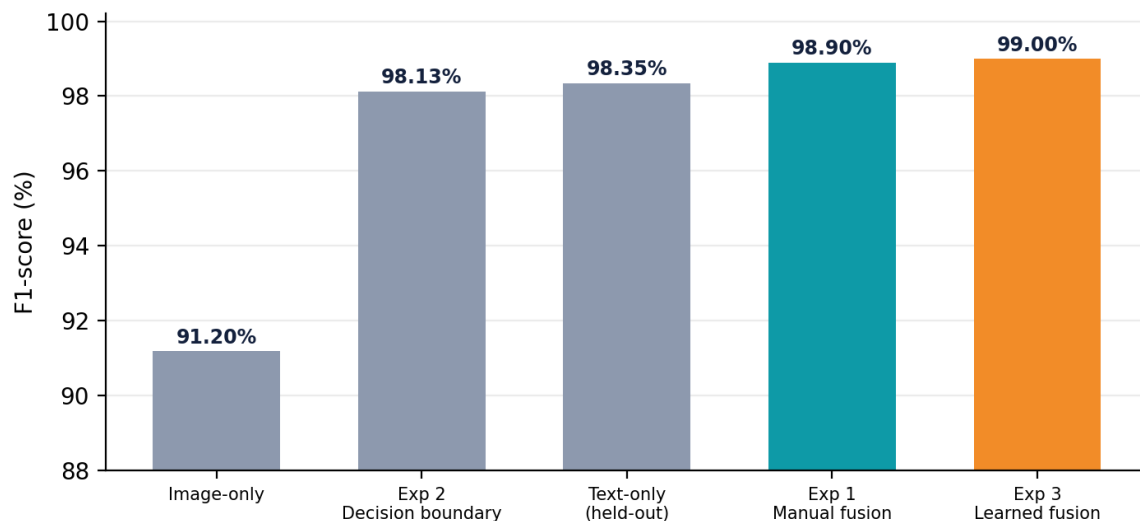


Figure 1 F1-score of every approach. Grey: baselines and the hard-boundary rule; teal: manual fusion; orange: final learned fusion. See the evaluation note in Section 9 on comparability.

## 1 Dataset and Pair Construction

| Item                         | Value                                                                       |
|------------------------------|-----------------------------------------------------------------------------|
| Product listings (train.csv) | 34,250 rows, 5 columns, grouped into 11,014 product groups (label_group)    |
| Unique images                | 32,412 (each embedded once with CLIP)                                       |
| Positive pairs               | 11,014 two listings sampled from each label group (seed 42)                 |
| Negative pairs               | 11,014 two random listings from different groups (NumPy generator, seed 42) |
| Total balanced pairs         | 22,028 (50% match / 50% non-match), shuffled with seed 42                   |
| Learned-fusion split         | Stratified 70 / 15 / 15: train 15,419 validation 3,304 test 3,305           |

Negative pairs are drawn at random, so most are easy to separate. This inflates absolute scores relative to a benchmark with mined hard negatives (see Limitations).

## 2 System Pipeline

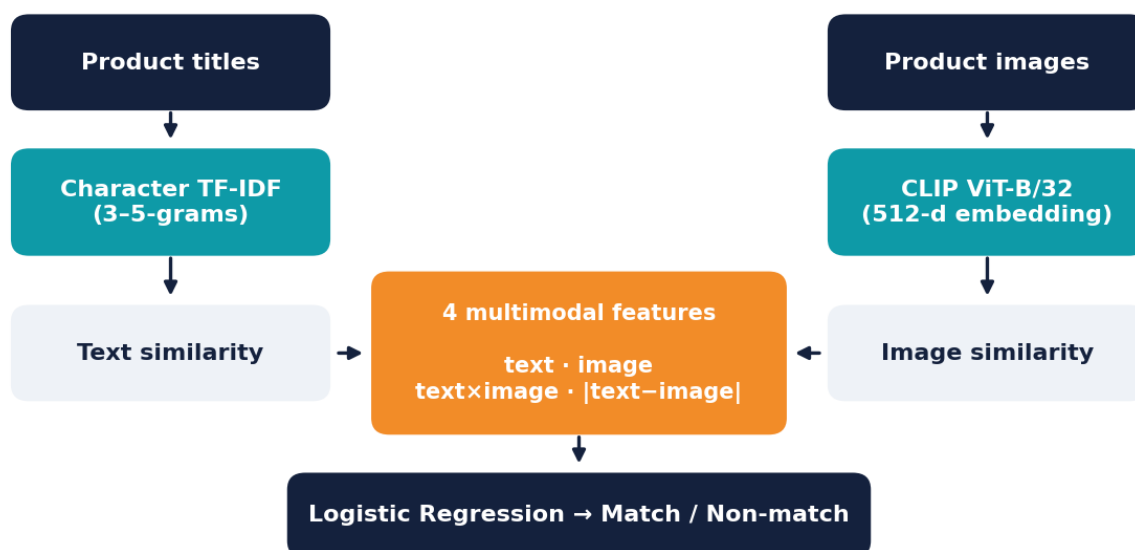


Figure 2 End-to-end multimodal matching pipeline.

The two feature extractors are deliberately simple and independent, which makes the fusion step easy to interpret.

| Component     | Configuration                                                                                                           | Output                                                  |
|---------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Text encoder  | TfidfVectorizer, analyzer = char, n-grams 3 5, min_df = 2, fitted on all pair titles                                    | Cosine similarity of title A and B (0 1)                |
| Image encoder | openai/clip-vit-base-patch32; images resized to 224 224; vision tower + visual projection; L2-normalised; batch size 32 | 512-d embedding per image; cosine similarity of A and B |
| Fusion        | Manual weighted sum, hard AND thresholds, or Logistic Regression                                                        | Match / non-match decision                              |

**Similarity statistics.** Text similarity separates the classes sharply (mean 0.510 for positives vs 0.006 for negatives). Image similarity separates them less cleanly (mean 0.789 vs 0.455, observed range 0.109 1.000), which already explains why text is the stronger signal.

**Compute cost.** Embedding all 32,412 images took 1,324 s (about 22 min, 1,013 batches) on the author's machine, far more than the text pipeline.

### 3 Experiments

#### Experiment 1 Manual Score-Level Fusion

**Hypothesis.** Text and image similarity are complementary, so a weighted combination  $S_{\text{fusion}} = S_{\text{text}} + (1 - \text{wt}) S_{\text{image}}$  should beat either modality alone. Five text weights were tried; for each, the threshold (0.00 1.00, step 0.01) was chosen to maximise F1.

| Text wt.    | Image wt.   | Threshold   | Precision     | Recall        | F1            |
|-------------|-------------|-------------|---------------|---------------|---------------|
| 0.00        | 1.00        | 0.59        | 91.81%        | 91.16%        | 91.48%        |
| 0.25        | 0.75        | 0.49        | 97.49%        | 95.91%        | 96.69%        |
| 0.50        | 0.50        | 0.36        | 99.10%        | 98.00%        | 98.55%        |
| <b>0.75</b> | <b>0.25</b> | <b>0.20</b> | <b>99.11%</b> | <b>98.70%</b> | <b>98.90%</b> |
| 1.00        | 0.00        | 0.06        | 98.72%        | 97.55%        | 98.13%        |

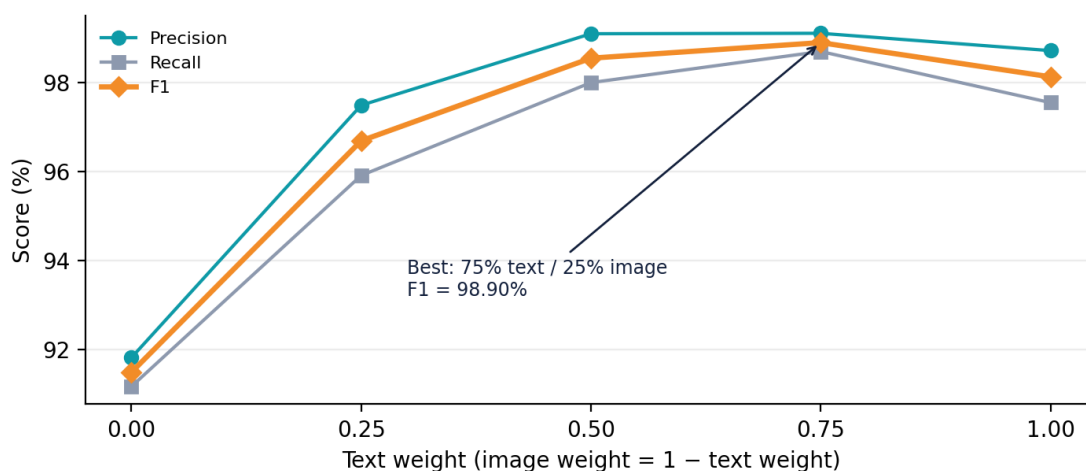


Figure 3 Precision, recall and F1 across text/image weightings (full balanced pair set).

The best mix is **75% text / 25% image** (F1 98.90% vs 98.13% text-only). Pushing image weight beyond 25% hurts, because CLIP tends to rate different products of the same broad category as similar.

**Rescue analysis.** Against the text-only rule (threshold 0.06), the fused rule corrected **209** pairs and broke **41**, a net gain of 168 pairs out of 22,028. Rescued true matches typically had very low title overlap but high image similarity (for example two ACOME charging-cable listings worded differently, text 0.045 / image 0.859). Rescued non-matches had moderate text overlap but weak imagery. Harmed pairs were mostly visually similar but different items (cosmetics, clothing, accessories), such as a phone case vs a hoodie-style listing, plus a few true matches whose photos differed greatly.

## Experiment 2 Multimodal Decision Boundary

**Hypothesis.** Separate thresholds, match if text  $X$  and image  $Y$ , might give a stricter, more precise rule. Text thresholds 0.00 0.20 and image thresholds 0.20 0.90 (step 0.01) were searched.

| Text threshold | Image threshold | Precision | Recall | F1     |
|----------------|-----------------|-----------|--------|--------|
| 0.06           | 0.20            | 98.72%    | 97.55% | 98.13% |

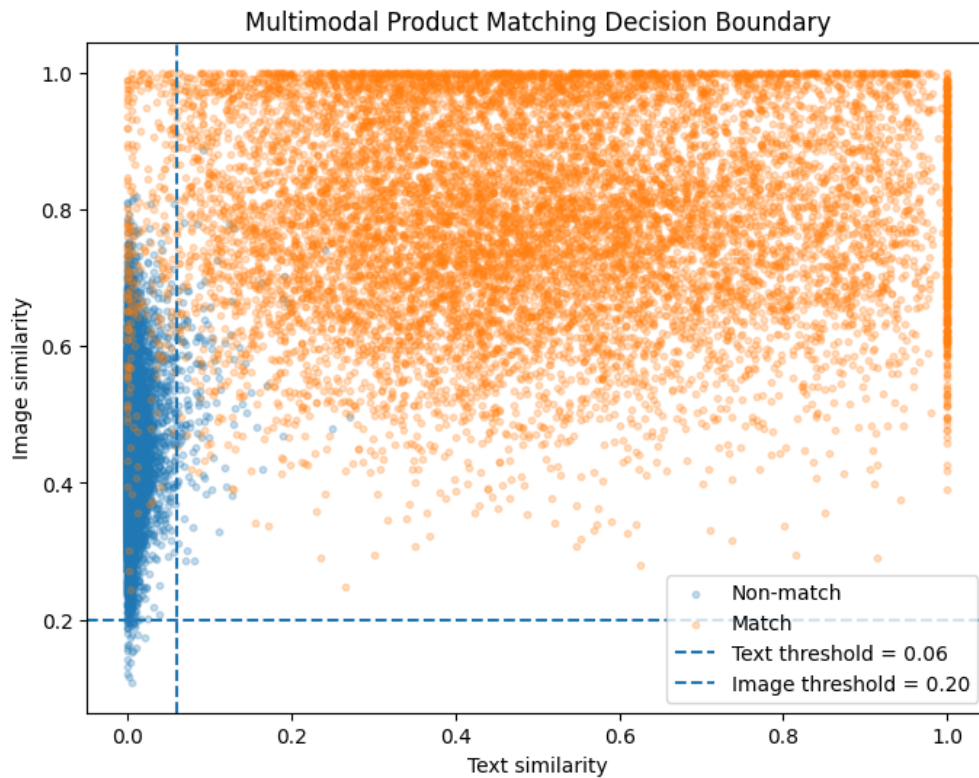


Figure 4 Text vs image similarity for all pairs, with the selected thresholds (from the notebook). Non-matches cluster at near-zero text similarity.

The optimal image threshold sat at the bottom of the search range because almost every pair already had CLIP similarity above 0.20; the image condition removed almost nothing and the result equals the text-only baseline. Raising it cut recall sharply. **Conclusion:** a hard AND rule does not improve on text-only matching; image similarity works better as soft evidence.

## Experiment 3 Learned Multimodal Fusion

**Hypothesis.** A model can discover a better combination of modalities than a fixed weighting. Four features were built for each pair:

| Feature                          | What it captures                                              |
|----------------------------------|---------------------------------------------------------------|
| text_similarity                  | Character-TF-IDF cosine similarity of the titles              |
| image_similarity                 | CLIP cosine similarity of the images                          |
| text_similarity image_similarity | Whether both modalities give strong evidence at the same time |
| text_similarity image_similarity | Disagreement between the two modalities                       |

A Logistic Regression (random\_state 42, max\_iter 1000) was fitted on the training split. The classification probability threshold was chosen on the validation split (0.01 0.99, step 0.01), giving **0.32**, and then applied once to the untouched test split.

| Metric    | Validation    | Held-out test |
|-----------|---------------|---------------|
| Threshold | 0.32          | 0.32          |
| Precision | 99.63%        | 99.39%        |
| Recall    | 98.73%        | 98.61%        |
| <b>F1</b> | <b>99.18%</b> | <b>99.00%</b> |

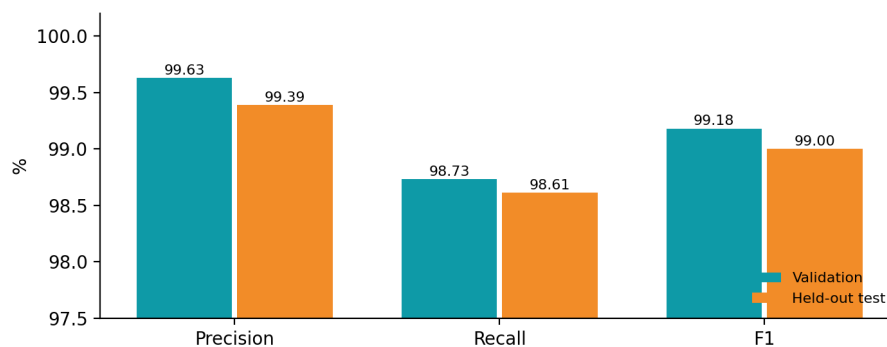


Figure 5 Validation vs held-out test performance of the learned fusion model. The small gap indicates little over-fitting.

### Learned coefficients (intercept 9.609):

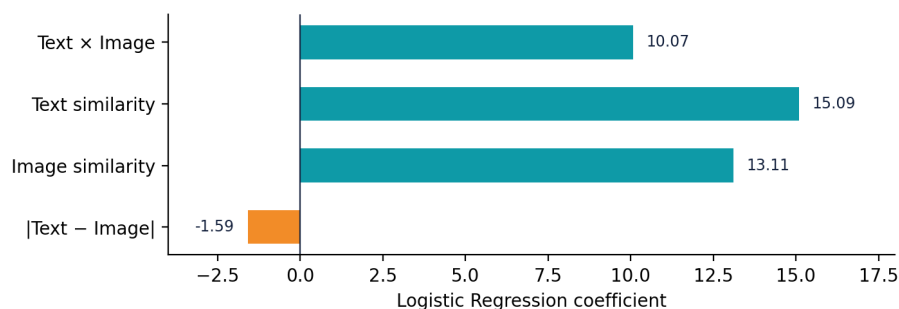


Figure 6 Logistic Regression coefficients on the four features.

Both similarities carry large positive weight (text 15.09, image 13.11) and so does their product (10.07), meaning agreement between modalities adds evidence beyond either alone. The disagreement feature has a small negative weight ( -1.59): when the signals diverge, the model becomes slightly more cautious. The gain over Experiment 1 is modest (98.90% → 99.00%), but the model uses just four features and stays fully interpretable.

## 4 Consolidated Results

| Approach                    | F1            | Key idea                                            | Protocol      |
|-----------------------------|---------------|-----------------------------------------------------|---------------|
| Text-only                   | 98.35%        | Character TF-IDF, threshold 0.06                    | Held-out test |
| Image-only                  | 91.20%        | CLIP similarity, threshold 0.62 (from validation)   | Held-out test |
| Exp 1 Manual fusion         | 98.90%        | 75% text + 25% image, threshold 0.20                | Full pair set |
| Exp 2 Decision boundary     | 98.13%        | Text 0.06 AND image 0.20                            | Full pair set |
| <b>Exp 3 Learned fusion</b> | <b>99.00%</b> | Logistic Regression over 4 features, threshold 0.32 | Held-out test |

## 5 Error Analysis

On the held-out test set (3,305 pairs) the final model made **10 false positives** and **23 false negatives**: 33 errors in total, or about 1.0%. Rather than reviewing all 33, representative cases were inspected visually.

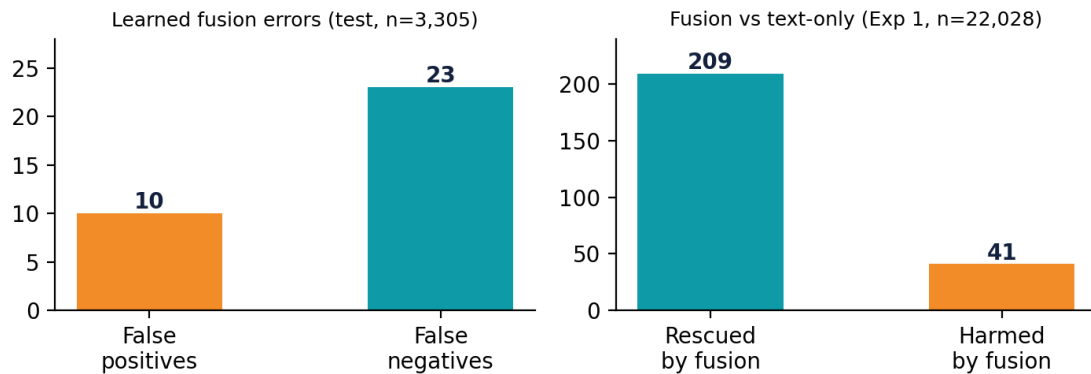


Figure 7 Left: final-model errors on the test split. Right: pairs rescued vs harmed by manual fusion relative to text-only (Experiment 1).

### False positives (model said Match, truth is Non-match)



Figure 8 Left to right, top to bottom: night cream vs face powder; Ara dress vs Hayla dress; storage bottles vs shoe rack; baby hangers vs spoon/fork pack.

## False negatives (model said Non-match, truth is Match)

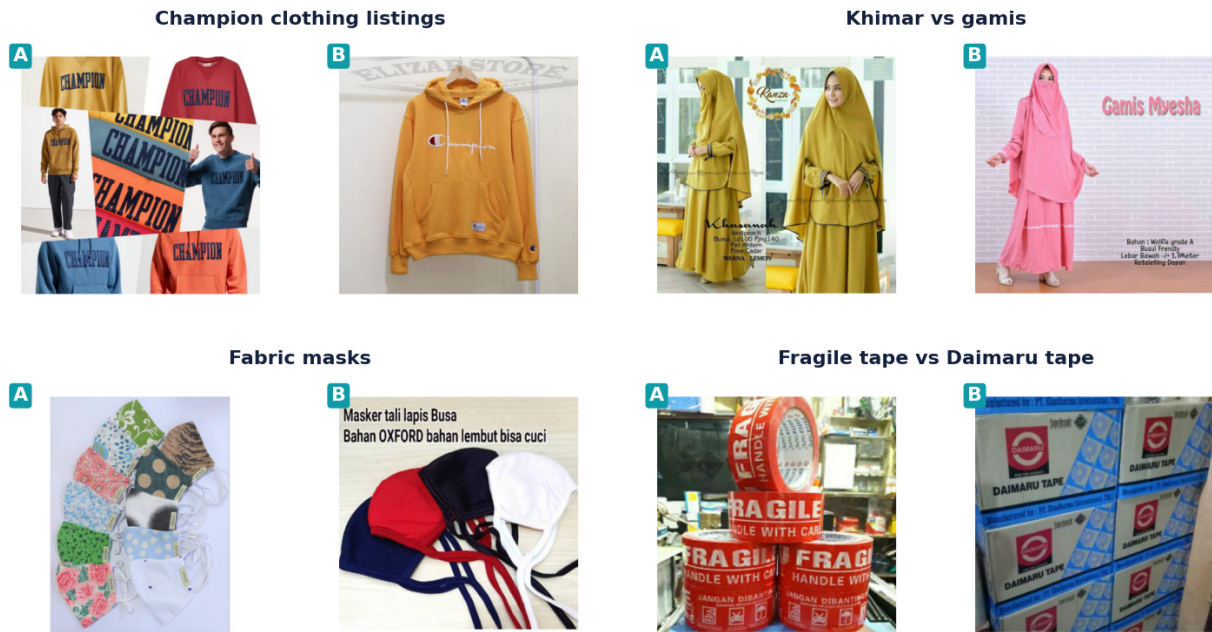


Figure 9 Champion clothing listings; khimar vs gamis; fabric masks; Fragile tape vs Daimaru tape.

| Case                            | Category               | Pred.     | Truth     | Why it failed                                                                                              | Potential fix                                         |
|---------------------------------|------------------------|-----------|-----------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Night cream vs face powder      | Visually similar       | Match     | Non-match | Similar small cosmetic containers; CLIP sees broad visual similarity while titles name different products. | Fine-grained visual features + stronger identity text |
| Storage bottles vs shoe rack    | Visually similar       | Match     | Non-match | Both are household storage items on clean backgrounds; encoder captures category and layout, not identity. | Fine-grained image matching                           |
| Baby hangers vs spoon/fork pack | Conflicting modalities | Match     | Non-match | Image similarity very high despite almost no textual evidence; visual signal dominated.                    | Better modality-disagreement handling                 |
| Champion clothing listings      | Noisy titles           | Non-match | Match     | Titles share almost no wording although images show related Champion clothing.                             | Semantic text embeddings / brand features             |
| Khimar vs gamis                 | Product variant        | Non-match | Match     | Same product family but different style, colour and presentation; both signals weak.                       | More robust semantic + visual representations         |
| Fragile tape vs Daimaru tape    | Different packaging    | Non-match | Match     | Same product type with different branding and packaging.                                                   | Category metadata + packaging-robust visual features  |

## Key observations

- **Visual overgeneralisation.** CLIP can treat different products as similar when they share category, shape, layout or photography style.
- **Textual over-reliance.** Character TF-IDF is strong with exact lexical evidence but fails on noisy or differently worded titles.
- **Conflicting modalities stay hard.** Fusion helps when one modality supplies missing evidence, but contradictory signals remain the main source of errors.

## 6 Ablation Study

Each modality was removed in turn. Text-only and image-only thresholds were selected on validation and evaluated on the same held-out test set.

| Model                                | Test F1       | Interpretation                                    |
|--------------------------------------|---------------|---------------------------------------------------|
| Image-only                           | 91.20%        | Useful visual signal, but weak for exact identity |
| Text-only                            | 98.35%        | Strongest individual modality                     |
| <b>Text + Image (learned fusion)</b> | <b>99.00%</b> | Best overall; image adds complementary evidence   |

Fusion improves F1 from 98.35% to 99.00%, a gain of **0.65 percentage points**. Because the baseline is already very strong, this equals roughly a 40% reduction in the remaining F1 error (1.65% → 1.00%). The image modality is valuable even though it is much weaker on its own.

## 7 Limitations

- **Text representation.** Character TF-IDF can miss semantic equivalence when titles use different words or languages.
- **Visual overgeneralisation.** CLIP can confuse different products that look broadly similar.
- **Variants.** Colour, size, packaging and model changes make exact identity difficult to judge.
- **Random negatives.** They are easier than mined hard negatives, so scores here are likely higher than on a harder benchmark.
- **Pair-level benchmark.** The task is binary classification of sampled pairs, not retrieval or ranking over a large candidate pool.
- **Compute.** Generating CLIP embeddings for all 32,412 images required substantial computation.
- **Label ambiguity.** Some listings in the same group have very different images or descriptions, which is hard even for a human to resolve.

## 8 Future Improvements

| Direction                        | Why it matters                                                       |
|----------------------------------|----------------------------------------------------------------------|
| Hard-negative mining             | Test and train on near-duplicate non-matches that look or read alike |
| Fine-tuned multimodal embeddings | Make embeddings discriminative for exact product identity            |
| Semantic text models             | Handle multilingual and noisy titles beyond character overlap        |
| Fine-grained visual models       | Separate products within the same visual category                    |
| Multi-stage retrieval            | Fast candidate generation followed by expensive reranking            |
| Retrieval evaluation             | Report Recall@K, Precision@K and mAP rather than pair F1 only        |
| Richer metadata                  | Add brand, category, price and colour as fusion features             |

## 9 Final Conclusion

The Finale shows that multimodal product matching benefits from combining complementary textual and visual evidence. Character-level TF-IDF is the strongest single signal, while CLIP contributes useful information where text is weak. The learned fusion model gave the best held-out result: **99.39% precision, 98.61% recall and 99.00% F1**, improving on the text-only ablation by 0.65 percentage points.

Multimodality works best when signals are treated as complementary evidence rather than as a strict requirement that both modalities independently agree.

**Evaluation note.** Experiment 3 and the ablation use a held-out train/validation/test protocol. The F1 values for Experiments 1 and 2 came from threshold searches on the full balanced pair set, so they are exploratory comparisons and not strictly like-for-like with the held-out numbers. The text-only baseline also differs by protocol: 98.13% on the full set, 98.35% on the test split.

## Reproducibility details

| Setting             | Value                                                                               |
|---------------------|-------------------------------------------------------------------------------------|
| Random seed         | 42 (pair sampling, shuffling, splits, Logistic Regression)                          |
| CLIP model          | openai/clip-vit-base-patch32, 512-d projected, L2-normalised embeddings             |
| Text vectoriser     | TfidfVectorizer(analyzer="char", ngram_range=(3, 5), min_df=2)                      |
| Threshold selection | Exp 1 2: full set   Exp 3 and ablation: validation split only                       |
| Libraries           | pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, Pillow, matplotlib |